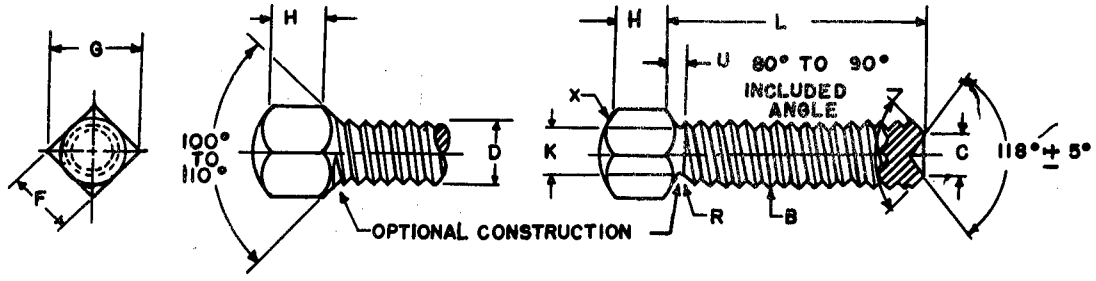
FED. SUP CLASS
5305



D	NOMINAL DIAMETER	1/4	5/16	3/8	7/16	1/2
B	THREADS PER INCH	20UNC-2A	18UNC-2A	16UNC-2A	14UNC-2A	13UNC-2A
K	DIAMETER OF NECK RELIEF Max	.185	.240	.294	.345	.400
	Min	.170	.225	.279	.330	.385
F	WIDTH ACROSS FLATS Max	.250	.3125	.375	.4375	.500
	Min	.240	.302	.362	.423	.484
G	WIDTH ACROSS CORNERS Min	.331	.415	.497	.581	.665
H	HEIGHT OF HEAD Nom	.187	.234	.281	.328	.375
	Max	.196	.245	.293	.341	.389
	Min	.178	.224	.270	.315	.361
R	RADIUS OF NECK RELIEF Max	.032	.036	.041	.046	.050
U	WIDTH OF NECK RELIEF Max	.100	.111	.125	.143	.154
X	RADIUS OF HEAD	.625	.781	.937	1.094	1.250
C	DIAMETER OF POINT Max	.132	.172	.212	.252	.291
	Min	.118	.156	.194	.232	.270

L LENGTH	Tolerance	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN
3/8	+0 -1/32	1		16							
1/2		2		17		34		51			
5/8		3		18		35		52		68	
3/4		4		19		36		53		69	
7/8		5		20		37		54		70	
1	+0 -1/16	6		21		38		55		71	
1-1/4		7		22		39		56		72	
1-1/2		8		23		40		57		73	
1-3/4		9		24		41		58		74	
2		10		25		42		59		75	
2-1/4	+0 -3/32			26		43		60		76	
2-1/2				27		44		61		77	
3				28		45		62		78	
3-1/2										79	
4										80	

MATERIAL: Steel, Carbon, in accordance with procurement specification.
HARDNESS: Setscrews shall be hardened by heat treatment to a hardness of Rockwell C30 to 37 or shall be surface hardened to not less than Rockwell 30N64.
PROTECTIVE COATING: Cadmium Plated, in accordance with Specification QQ-P-416, Type II, Class 3.
Zinc Plated, in accordance with Specification QQ-Z-325, Type II, Class 3.
THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services Handbook H-28.
DIMENSIONS: All dimensions are in inches unless otherwise specified.
PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS51054-1.
NOTES: (1) Referenced documents shall be of the issue in effect on date of invitations for bid.
(2) This standard takes precedence over documents referenced herein.

Ⓐ CANCELLED
SEE SHEET 2

This document has been promulgated by the Department of Defense as the Military Standard to limit the selection of the item, product, or design covered herein in engineering, design, and procurement. This standard shall become effective not later than 90 days after the latest date of approval shown.
Ⓐ For changes see sheets 1 and 2

APPROVED 7 MAY 58 REVISED Ⓐ 18 NOV 63

P.A. WC Other Cust SHIPS ASD	TITLE SETSCREW-SQUARE HEAD, CUP POINT, CARBON, STEEL, CADMIUM OR ZINC, UNC-2A	MILITARY STANDARD MS51054	
PROCUREMENT SPECIFICATION FF-S-103	SUPERSEDES:	SHEET 1 OF 2	

DD FORM SEP 57 672-1 (Coordinated) Previous editions of this form are obsolete.

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

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FED. SUP CLASS 5305

D	NOMINAL DIAMETER	5/8	3/4	7/8	1	
B	THREADS PER INCH	11UNC-2A	10UNC-2A	9UNC-2A	8UNC-2A	
K	DIAMETER OF NECK RELIEF Max	.507	.620	.731	.838	
	Min	.492	.605	.716	.823	
F	WIDTH ACROSS FLATS Max	.625	.750	.875	1.000	
	Min	.606	.729	.852	.974	
G	WIDTH ACROSS CORNERS Min	.833	1.001	1.170	1.337	
H	HEIGHT OF HEAD Nom	.468	.562	.656	.750	
	Max	.485	.582	.678	.774	
	Min	.452	.544	.635	.726	
R	RADIUS OF NECK RELIEF Max	.059	.065	.072	.081	
U	WIDTH OF NECK RELIEF Max	.182	.200	.222	.250	
X	RADIUS OF HEAD	1.562	1.875	.2187	2.500	
C	DIAMETER OF POINT Max	.371	.450	.530	.609	
	Min	.347	.425	.502	.579	

L LENGTH	Tolerance	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN
3/4 7/8	+0 -1/32	86 87									
1 1-1/4 1-1/2 1-3/4 2	+0 -1/16	88 89 90 91 92		104 105 106 107 108		120 121 122 123		136 137 138			
2-1/4 2-1/2 3 3-1/2 4 4-1/2 5	+0 -3/32	93 94 95 96 97 98		109 110 111 112 113 114		124 125 126 127 128 129 130		139 140 141 142 143 144 145			

Ⓐ CANCELLED SEE NOTE

NOTE: For material, protective coating and other pertinent data, sheet 1.

Ⓐ CANCELLED AFTER 18 November 1963. Use MS 51955

INTERCHANGEABILITY RELATIONSHIP - MS 51955 can replace the cancelled screws of the same size. The cancelled MS51054 cannot always replace MS 51955 and should be used until existing stock is depleted. Then use only the superseding MS screws of the same size for design and replacement.

APPROVED 7 MAY 58 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC Other Cust SHIPS ASD	TITLE SETSCREW- SQUARE HEAD, CUP POINT, CARBON STEEL, CADMIUM OR ZINC, UNC-2A	MILITARY STANDARD MS51054	
PROCUREMENT SPECIFICATION FF-S-103	SUPERSEDES:	SHEET 2	OF

DD FORM 1 SEP 57 672-1 (Coordinated) Previous editions of this form are obsolete.

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